

Walk-in runbook

The defence is a cold run on a space nobody here has seen, with the examiners choosing the tier and measuring the room with a laser while it runs. This page is what to do on the day.

Everything below is current as of the latest commit. If a number here disagrees with [bench/results/gates.json](#), **the JSON is right and this page is stale** — it is generated by [bench/gates.py](#) and this page is written by hand.

Before leaving

```
cd cozmo-ai-assignment && source .venv/bin/activate
pytest -q # 98 passed
scanplan --version # scanplan 0.1.0
```

The LiDAR tier needs no weights and no network. That is deliberate and it is why this section is three lines. If there is any chance the examiners pick video or photo, also run:

```
pip install "torch>=2.4,<2.10" "torchvision>=0.19,<0.25" \
  --index-url https://download.pytorch.org/whl/cpu # CPU wheels; the default pulls CUDA
pip install -e ".[dev,models]" # then transformers + a 95 MB checkpoint
python -c "from scanplan.ingest.video import _depth_model; _depth_model()" # pre-cache weights
```

Without that extra, the video and photo tiers raise a stated error naming the exact command rather than a traceback — survivable, but a download at the table is not.

Step 0, always: ask what it is before measuring it

```
scanplan inspect /path/to/capture
```

This prints which tier was detected and what the pipeline thinks it was handed — frame count, duration, whether the walk closed its loop, whether tracking relocalised. Thirty seconds here prevents measuring the wrong thing confidently. If the detected tier is not the one the examiners chose, stop and sort out the folder before running.

If they choose LiDAR

1. They capture with **Stray Scanner**, following [docs/capture_protocol.md](#).
2. Export the folder from the phone, copy it across.
- 3.

```
scanplan run /path/to/export
```

4. Expect **roughly 12-57 s** depending on walk length, against a 60 s budget. Read the room table off [summary.md](#), or open [plan.svg](#) to show the plan. [result.json](#) is the contract document.

Know this before the day: the margin is thin. The longest capture measured 56.5 s of a 60 s budget on an idle 12-core machine. A slower laptop will blow it. If that happens, say so plainly rather than hiding it — and `--stride 4` roughly halves the time at a small accuracy cost, which is a legitimate thing to do in front of examiners if you announce it.

The honest footnote: adaptive keyframe selection would have brought the worst capture to 40.8 s with real headroom. It was declined because its threshold turned out to be the only value in the swept space that cleared three gates at once ([docs/declined_changes.md](#) §1). This thin margin is the price of that decision, and it is the right price to have paid.

This is the strongest tier and the one to steer towards if given a choice. It is also the only tier with a metric scale that was *measured* rather than inferred.

If they choose video

Built and honest about its limits. It needs **RGB plus a pose track** — [rgb.mp4](#) and [odometry.csv](#) in one folder:

```
scanplan run /path/to/folder # or point at the .mp4 directly
```

Say this plainly before the number appears: **the video tier ignores depth entirely** and infers it with a metric monocular model, so it is a genuine measurement of what is lost when the depth sensor goes away. On the supplied captures its footprint lands up to **60% away** from the LiDAR result on the same walk —

G-WALL-VIDEO is NOT MET and is reported as such.

What *is* met is **A-CALIB-VIDEO**: the intervals are widened $\times 11$ and the LiDAR reference falls inside them on 2 of 2 captures. The claim is not "this is accurate", it is "this says how wrong it might be, and it is right about that."

If handed a bare `.mov` with no poses, it raises a stated error. Estimating poses from video alone is structure-from-motion and is not implemented — do not improvise.

If they choose photo

Built, and **fails one gate by construction rather than by accident**. Input is a folder of per-room folders of stills; focal length comes from EXIF.

```
scanplan run /path/to/rooms_folder
```

Each room returns as the bounding box of what one view could see. **G-PHOTO-STITCH is NOT MET: the rooms come back as disconnected groups**. With no camera poses, nothing in the input says how the rooms relate — an unposed stills set does not contain the answer to "which wall do these two rooms share". The output says so in a `PHOTO_TIER_NOT_STITCHED` error-severity warning rather than laying out a plausible arrangement.

Also state: an L-shaped room returns as a rectangle. The tier reports room boxes, not outlines.

What to say about the numbers

- Every value carries a **nominal 90% interval**. Quote the interval, not just the value. The schema enforces `ci_low ≤ value ≤ ci_high`; **A-SCHEMA** is 3/3 valid across the three tiers.
- **Depth bias is measured, not assumed**. The device reads **18.0 mm short** of FARO laser truth — pooled median over 4.79 million pixels across 8 ARKitScenes scans where laser depth is registered to the same frames. It is corrected at ingest. Per-scan medians ran -14 to -27 mm, so the **4.1 mm standard deviation** is what the single correction leaves unexplained and that is carried on every surface height.
- **Ceiling height** is the tightest quantity: on synthetic geometry whose true height is 2.500 m by construction, it reads **2.497 m — 3.1 mm low, σ 5.7 mm**, and the floor lands within 1.5 mm of zero. Measured through `scanplan run` with the shipped defaults, not a configuration only the tests use.
- **Floor area is measured, not filled** — it is the floor actually covered, so it can only under-report, never invent. On the synthetic room it reads **-0.05%**. An early version that flood-filled came out 3 $\times$ too large, which is the failure this design removes.
- **Room count may not match what a human would say**. It under-splits. A merged pair still reports a correct combined area; an invented room does not.
- **Runtime is not in `gates.json`** on purpose — wall-clock is a property of the machine. It is in `bench/results/timing.json`, which is why that one file is expected to differ on a re-run and the other benchmarks are not.

Gates: what passes, what does not, and what is unmeasured

Have this straight before being asked. Full table in `docs/compliance_matrix.md`, generated from `gates.json`.

	Gate	State
✓	A-RUNTIME, A-DET, A-SCHEMA, G-DRIFT	met
✓	G-REPEAT-ROOMS	met — 5 rooms vs 5 from two walks of one flat
✓	A-CALIB-VIDEO	met — reference inside the widened interval, 2/2
✓	G-H2H-ENGINEER	met — 9/10 (90%) vs an independent implementation on exact truth
✗	G-REPEAT-FOOTPRINT	not met — 3.2% apart
✗	G-WALL-VIDEO	not met — 60% worst
✗	G-PHOTO-STITCH	not met , by construction
✓	G-CEIL	met — 5/5 walks within 15 mm of FARO laser truth, mean -4.3 mm
✗	G-CEIL-SPREAD	not met — 10.3 and 12.0 mm against a 10 mm target
✗	G-OPEN	not met — 0/12 openings agree within 2 cm between two walks

	Gate	State
✓	A-WALL-LIDAR	met — 10/10 at 8.6 mm under a <i>stated narrowing</i> : the distance between the walls we name. 8/10 if naming is also charged — quote that if asked
✗	G-WALL-PHOTO	not met — 0/6, median 75%; no poses means one view's reach bounds the box
○	G-REPEAT	not measurable on this pair, with evidence
○	G-H2H	pending — needs tape truth that does not exist
○	A-DMG-DETECT	not built

The two repeatability gates are the interesting answer: **exactly one of the two passes, and the depth-bias correction flipped which one**. Before the correction it was 4-vs-5 rooms (failing) with 0.7% footprint agreement (passing). That is reported, not absorbed — two gates that trade places under an 18 mm sensor correction were never measuring two independent properties. [docs/fix_loop.md](#) §10 has it in full.

If something goes wrong

Symptom	Do this
<code>error: ... no floor found and "none of the sampled frames looked below the camera"</code>	Not a bad capture — a periodic sweep aliasing with the frame stride. Re-run with <code>--stride 1</code> . This is a real failure mode: one capture in hand cycles its pitch every 3 frames and the default stride is 3
<code>error: ... no floor found</code> (points are below the camera)	The capture genuinely never showed enough floor. Ask for a re-walk with the floor sweep in capture_protocol.md §Tier 1 step 4
<code>error: ... incomplete Stray Scanner export, missing ...</code>	Wrong folder level — point at the one containing <code>odometry.csv</code>
<code>error: the video and photo tiers need the model extra</code>	<code>pip install -e ".[dev,models]"</code> . The LiDAR tier still works; offer it
<code>error: ... the video tier needs a pose track</code>	A bare video carries no poses. Not recoverable on the day
A frame fails to decode	Already handled: skipped with a warning, the run continues. One bad frame in nine thousand must not cost a plan
Tier detected wrongly	<code>scanplan inspect</code> first, fix the folder layout, do not force it
Anything unexpected	The CLI prints one line and exits non-zero, never a traceback. <code>SCANPLAN_TRACEBACK=1</code> gives the full trace for debugging

Questions to expect, and the short answers

"Why no ML?" The LiDAR tier does not need it — the phone measures distance directly, so a model could only add a failure mode and a download. Models are *required* for video and photo, where a plain camera measures nothing and scale is mathematically unobservable: a dollhouse and a room project identically. Both those tiers use a metric depth model for exactly that reason. Nothing in the LiDAR path that produces a number uses a model.

"Where is your ground truth?" Two measurements are against a surveying instrument. The **depth bias** (18 mm per-pixel over 4.79 M pixels) and **ceiling height** — 5 of 5 ARKitScenes walks within 15 mm, mean −4.3 mm, with a leave-one-venue-out correction so no walk is corrected with its own truth. If asked how the correction is justified: uncorrected the same walks give 1 of 5 and −22.3 mm, so the ablation shows the correction doing the work. Everything else is either synthetic geometry we built and therefore cannot be wrong about, or one tier against another. There is no truth for the supplied captures — nobody here has stood in that property — and **G-CEIL** reports **NOT MEASURED** rather than a number it cannot defend. Three approaches to measuring it are written up in [bench/ceiling_vs_laser.py](#), all failing for the same reason: ARKitScenes venues are overwhelmingly multi-storey, and storey height needs one floor and one ceiling.

"Your intervals look wide." They carry the 4.1 mm the measured bias correction leaves unexplained. Without a bias term the ceiling σ computes to 0.1 mm, which is the standard error of 250,000 points and is nonsense — averaging removes noise, not a systematic offset.

"Why is your room count wrong?" It under-splits. `door_max_m` is one global number and rooms are not one size: wide enough to separate a room from a hallway erases a small bathroom. Two attempts at this are documented in [docs/fix_loop.md](#), the second reverted. The honest conclusion is that it is not a parameter problem.

"Does the benchmark reproduce?" `bash bench/clean_clone_check.sh` — fresh clone, fresh install, regenerate, diff. With `--with-models`, **eight of eleven come back byte-identical** and the other three are

named along with why they cannot: two are historical snapshots of code states that no longer exist, one is wall-clock. It has caught five real defects, including that `fix_loop_diagnosis.py` was itself measuring a pipeline we do not ship.

"Has anyone else's implementation been compared against yours?" Yes, and it is a committed gate. `cozmo-scan`, an independent submission to this same brief, reads the same public capture format, so both pipelines were handed the **identical** synthetic capture of a room that is 4.00 × 3.00 m by construction. **Beat or tie on 9 of 10 dimensions, 90% — G-H2H-ENGINEER.** The one loss is ceiling height on a capture modelling an *unbiased* sensor, where our 18 mm correction is unwarranted. Two captures are run on purpose, one modelling the device as measured and one modelling an ideal sensor, because running only the one that suits our correction would be choosing the input. This does **not** satisfy Part 3, which asks for a consumer scanning app; that is `magicplan`, and it is still PENDING.

"Where does their implementation beat yours?" Answer this one straight, because the benchmark already does and being caught hedging would be worse than the losses themselves. Three places, all measured:

- 1. Room splitting.** They report 3, 8 and 9 rooms on the three supplied captures where we report 2, 5 and 5. We under-split, it is a documented failure mode, and it is the gate our fix loop declared, predicted wrong, attempted twice and reverted. An independent implementation does it better.
- 2. Ceiling height per room.** They fit one per room — nine distinct values from 2.29 to 3.10 m on `c7d28f72c6` — where we fit a single storey height, 3.087 m for every room. No truth exists, so neither is provably right, but a uniform 3.09 m across a flat is on the high side and their model describes a real property more closely.
- 3. Frame selection.** They pick keyframes adaptively by pose change; we use a fixed stride. That is exactly why `scanplan run` failed on a capture whose sweep aliased with our stride and theirs did not. We shipped a `--stride` escape hatch and a diagnostic, which is a mitigation, not a fix.

What stays ours: **dimensional accuracy against exact truth** (the gap is large, consistent across both generators, and the convention is matched — their source specifies inside faces as ours does), and the **measured depth bias**, which they carry as a 13 mm uncertainty rather than correcting.

"Did you reject anything that would have improved a number?" Yes, four things, and they are written up in `docs/declined_changes.md` with their measured outcomes. The one worth naming: adaptive keyframe selection improved runtime 52.7 → 40.8 s and fixed a real aliasing failure, but its threshold of 2.9° turned out to be the *only* value in the swept space that clears three gates at once — the value its own stated derivation gives leaves G-REPEAT-ROOMS not met. That is a constant chosen by its effect on the score, so it did not ship. Also declined: a wall-plane change that reached 11/11 MET only by letting the device choose which surfaces the laser then measured, which relaxes what the gate penalises.

"What would you do with another week?" Find a single-storey ARKitScenes venue with its laser cloud and close G-CEIL; replace the global `door_max_m` with a split that adapts to local room scale; get tape truth into the head-to-head. In that order — the first turns an unmeasured gate into a measured one, which is worth more than improving a number already known.

The one thing not to do

Do not improvise a number. Every tier either measures or says it cannot: a stated NOT MEASURED is worth more than a confident wrong plan, and the whole submission is built on that. If asked for something the pipeline does not produce, say it does not produce it.